

Trainings ▾

Initial Check

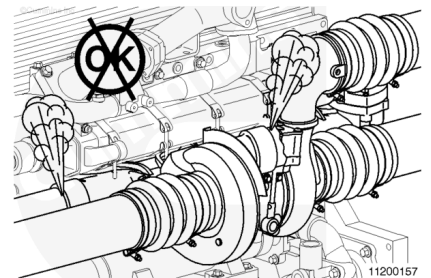
With CM870 and CM570

Operate the engine and inspect the exhaust manifold for leaks at the slip joints.

If the manifold is leaking at the slip joints, it **must** be repaired.

Some ISM engines are equipped with exhaust seal rings located within the slip joints. All ISM CM876 engines are equipped with exhaust seal rings.

Note : For engines not equipped with seal rings, the kit for the front slip joint is different than the kit for the rear slip joint. The outer shells are different, as well as the graphite seal. Only the clamps are similar.



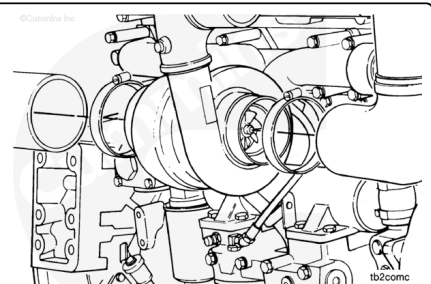
Repair

Automotive with CM875, CM870 and CM570

⚠ WARNING ⚠

To reduce the possibility of personal injury from hot surfaces, allow the engine to cool before beginning work. Wear appropriate hand and eye protection.

Note : For engines without exhaust seal rings.



Exhaust Manifold Slip Joint Seal Kit Contents

Part Description	Quantity per Kit
Covers	2
Seal	1
Clamp	1

Exhaust Manifold Slip Joint Seal Kit Contents

Part Description

Quantity per Kit

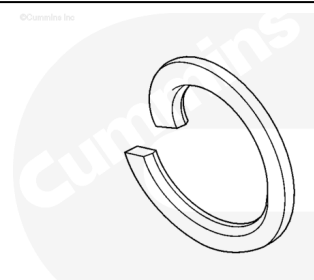
Note: Each kit contains parts for one slip joint repair. Two kits, front and rear, will be needed to service the front and back.

Remove the turbocharger inlet and exhaust piping. Refer to Procedure 010-033 in Section 10.

(/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)

Remove the front and rear exhaust manifold heat shields located below the manifold. Refer to Procedure 011-032 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-032-tr.html)

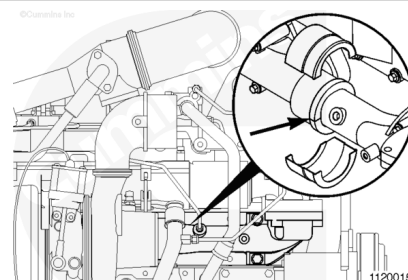
Twist the graphite seal to open it, and place it on the end section of the manifold, up against the slip joint.



Note : The recessed inside diameter goes against the center section. The split in the seal is located horizontally away from the engine.

Place the top and bottom shells over the graphite ring and the exhaust housing center section step.

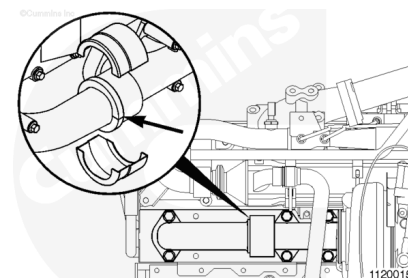
When placing the shells over the seal, start with the back edges closer together than the front. When the clamp is tightened, it will pull the graphite split tighter together.



Place the top and bottom shells over the slip joint and graphite seal with the notch in the cover toward the center of the exhaust manifold center section.

Note : A pry bar can be used to get the outer shells pressed together enough to get the clamp on. Do not damage the EGR cooler with the pry bar.

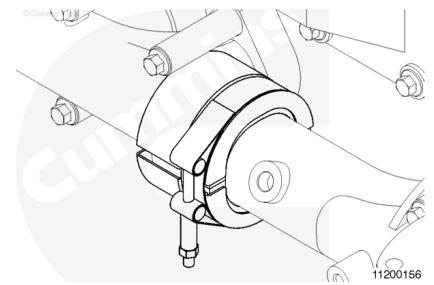
Be sure to keep the mating edges aligned with the exhaust manifold parting line.



Front Seal - Note the location and orientation of the clamp.

Tighten the clamp.

Torque Value: 11 n•m [97 in-lb]



Rear Seal - Note the location and orientation of the clamp.

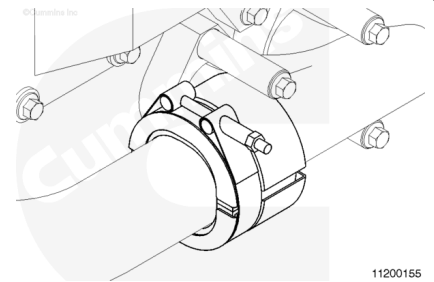
Tighten the clamp.

Torque Value: 11 n•m [97 in-lb]

Replace the heat shields. Refer to Procedure 011-032 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-032-tr.html>)

Replace the turbocharger plumbing.

Clean the soot from around the slip joint area.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Automotive with CM870

- Disconnect the batteries. See equipment manufacturer service information.

- Drain the coolant. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Remove the air piping from the turbocharger.
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)
- Remove the EGR valve coolant lines. Refer to Procedure 011-022 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-022-tr.html)
- Remove the water manifold return tube. Refer to Procedure 008-061 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-061-tr.html)
- Remove the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)
- Remove the EGR valve. Refer to Procedure 011-022 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-022-tr.html)
- Remove the exhaust gas pressure sensor tube. Refer to Procedure 011-027 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-027-tr.html)

Automotive with CM876

Note : Brush away any loose dirt from the vicinity of the air handling connections to avoid contamination of the interior of the engine.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the coolant. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Remove the air piping from the turbocharger. Cover open points using caps from the Air Handling and Vehicle Air Plumbing Clean Care Kits.
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)
- Remove the water manifold return tube. Refer to Procedure 008-061 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-061-tr.html)
- Remove the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)
- Remove the exhaust gas pressure sensor tube. Refer to Procedure 011-027 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-027-tr.html)

Automotive with CM570

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the air piping from the turbocharger.
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)

Marine Applications

- Disconnect the batteries. See equipment manufacturer service information.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)
- Remove the heat exchanger. Refer to Procedure 008-053 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-053-tr.html)
- Remove the aftercooler assembly (sea water). Refer to Procedure 010-005 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-005-tr.html)

Remove

Remove the EGR cooler coolant return tube bracket from the exhaust manifold.

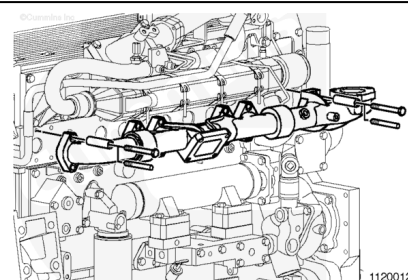
Remove the exhaust manifold.

Cover all open points on the exhaust manifold and cylinder head with the appropriate protective caps or heavy tape, if needed.

Remove the two upper capscrews from the end sections of the exhaust manifold and install two guide studs, Part Number 3376488.

Remove the remaining capscrews, exhaust manifold, and gaskets.

Discard the gaskets.

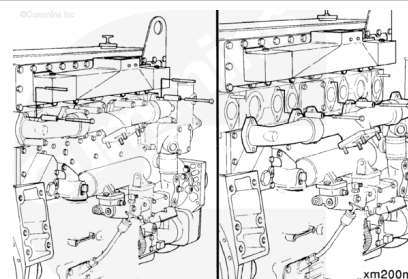


Automotive with CM570

Remove the two upper capscrews from the end sections of the exhaust manifold and install two guide studs, Part Number 3376488.

Remove the remaining capscrews, exhaust manifold, and gaskets.

Discard the gaskets.



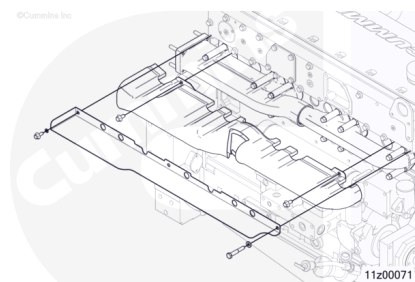
Marine Applications

Remove the heat shield mounting capscrews.

Remove the heat shield.

Remove the insulation blanket mounting capscrews.

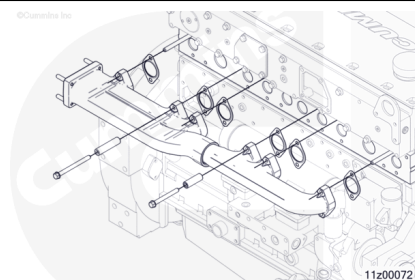
Remove the insulation blanket.



Remove an upper capscrew and spacer from each end of the exhaust manifold and install guide studs, Part Number 3376488.

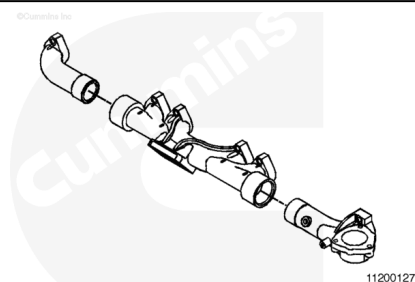
Remove the remaining capscrews, spacers, exhaust manifold, and gaskets.

Discard the gaskets.

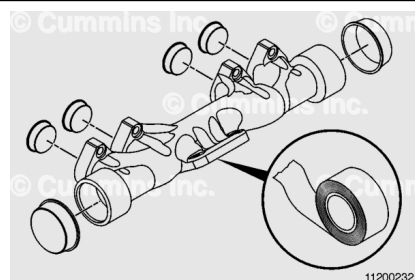


Disassemble

Remove the exhaust manifold end sections.



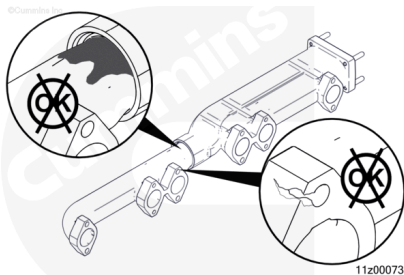
Cover the open points of the exhaust manifold with the appropriate protective caps or heavy tape, if needed.



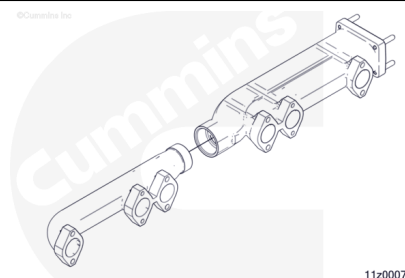
Marine Applications

Inspect the exhaust manifold slip joint for signs of soot leakage.

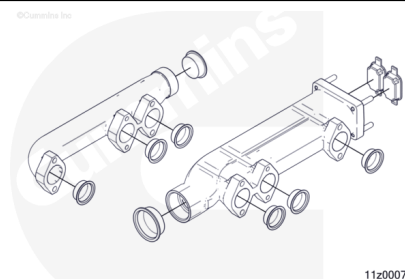
If there is no evidence of an exhaust leak from the slip joint and there are no visible cracks or damage, do **not** separate the exhaust manifold sections.



If soot leakage or damage is found, separate the manifold sections.



Cover the exhaust manifold openings with protective caps or heavy tape.



Clean and Inspect for Reuse

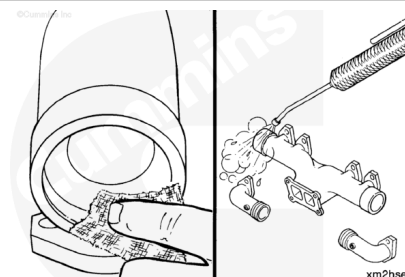
Automotive with CM875, CM870 and CM570

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Use 240-grit emery cloth to remove carbon deposits from the sealing surfaces.

Use steam to clean the exhaust manifold.

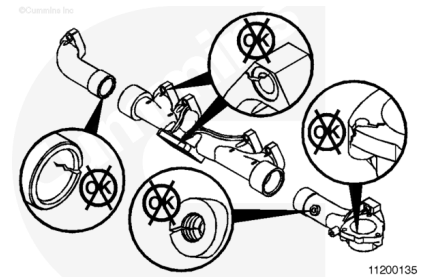
Dry with compressed air.

Inspect the parts for cracks or other damage.

A charge air cooler failure can cause progressive damage to the exhaust manifold. If the exhaust manifold is damaged, check the charge air cooler. Refer to Procedure 010-027 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-027-tr.html>)

Inspect the exhaust manifold slip joint for wear.

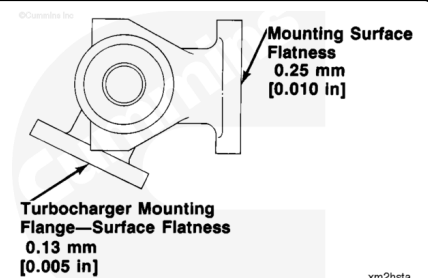
Inspect the exhaust manifold turbocharger mounting surface for warping.



Inspect the mounting surfaces for flatness.

The exhaust manifold mounting surface **must** be flat to within 0.25 mm [0.010 in]. The turbocharger mounting flange **must** have a surface flatness of 0.13 mm [0.005 in].

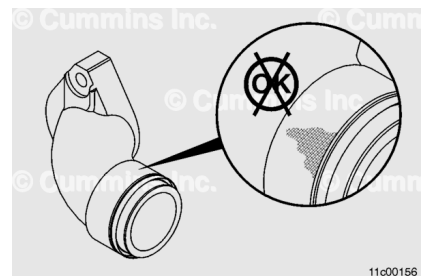
If these surfaces are **not** within the flatness specifications, see the Alternative Repair Manual, L10 and M11 Series Engines, Bulletin 3810310.



Automotive with CM876

Inspect the exhaust manifold slip joints for signs of soot leakage.

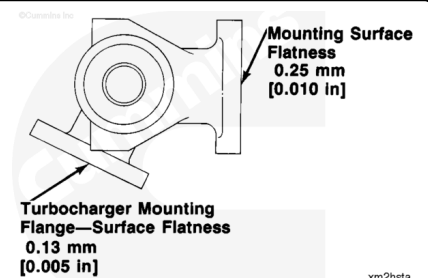
If there is no evidence of an exhaust leak from the slip joints and there are no visible cracks or damage, do **not** remove the exhaust manifold end sections.



Inspect the exhaust manifold turbocharger mounting flange for warpage.

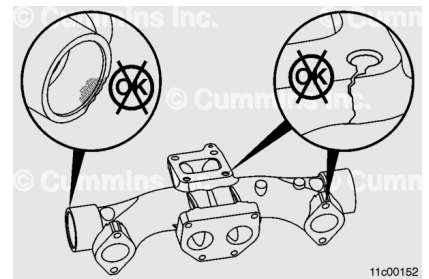
The exhaust manifold mounting surface **must** be flat to within 0.25 mm [0.010 in].

The turbocharger mounting flange **must** have a surface flatness of 0.13 mm [0.005 in].

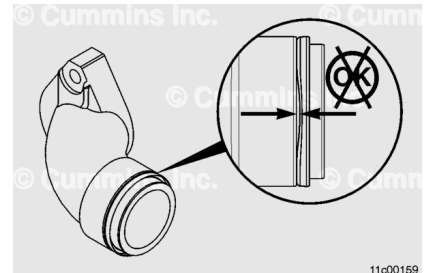


Inspect for cracks or other damage.

Note : A charge air cooler failure can cause progressive damage to the exhaust manifold. If the exhaust manifold is damaged, check the charge air cooler. Refer to Procedure 010-027 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-027-tr.html>)



Check that the exhaust seal is fully seated against the shoulder of the exhaust manifold male end section. If the exhaust seal is **not** fully seated against the shoulder, the exhaust seal **must** be replaced.

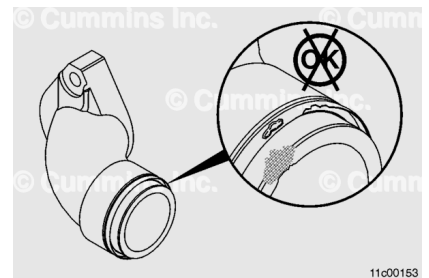


Inspect the manifold seal bore for wear.

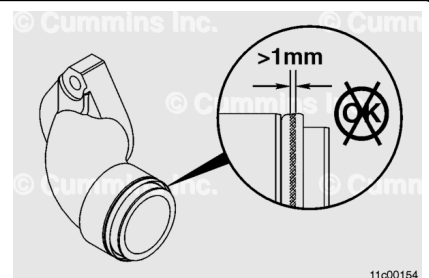
Inspect the exhaust seal for signs of damage or excessive wear.

The exhaust seal **must not** show signs of dents or other damage.

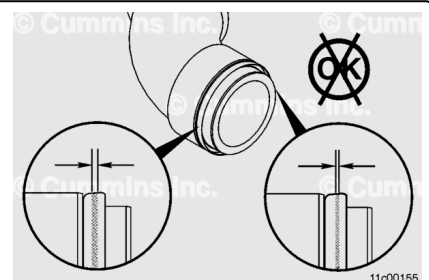
Measure the wear width on the exhaust seal ring. The wear width **must not** be more than 1 mm [0.04 inch] wide anywhere on the seal.



Wear width larger than 1 mm [0.04 in] indicates a failure and the exhaust seal **must** be replaced.

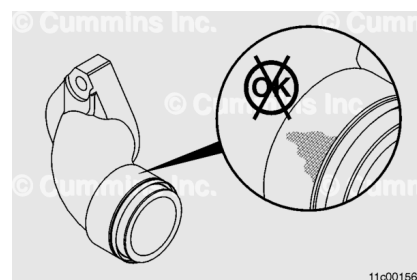


The wear width **must** be uniform around the seal. If one side of the exhaust seal ring is worn more than the other, the exhaust seal **must** be replaced.



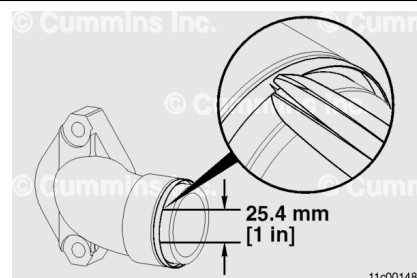
If there is no evidence of an exhaust leak from the slip joints and there are no visible cracks or other damage, do **not** remove the exhaust manifold seal rings.

If there is a failure or damage, remove the exhaust manifold seal rings.



Remove the exhaust seal rings, located within the slip-joint.

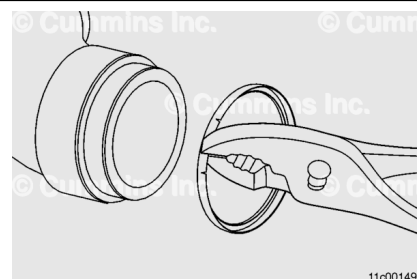
Use tin snip or side cuts, to cut halfway through the seal in two locations, about 25.4 mm [1 in] apart.



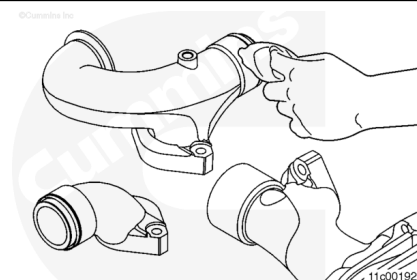
Twist and pull the seal off the exhaust manifold end sections.

Take care **not** to damage the seal carrier surface.

Discard the exhaust seal ring.

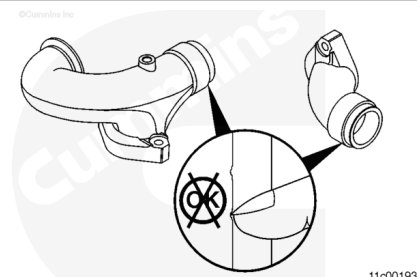


Use a soft cloth or a shop towel wetted with a safety solvent to remove carbon deposits from the sealing surfaces.

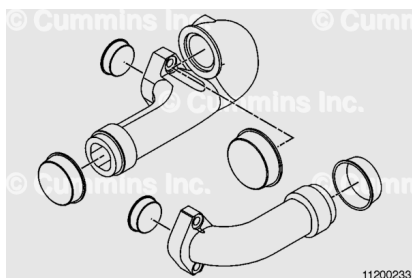


Inspect the exhaust seal ring mating surfaces of the male and female sections of the exhaust manifolds.

If scratches or nicks can be felt with a fingernail on the exhaust seal ring mating surfaces, the exhaust manifold section **must** be replaced.



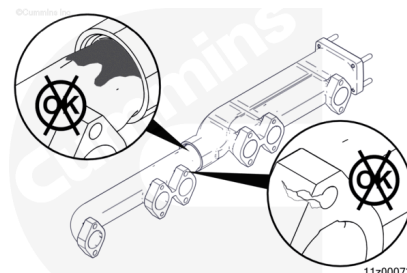
After completing inspections, cover the exhaust manifold with the appropriate protective caps or heavy tape, if needed.



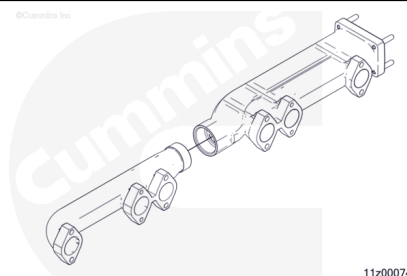
Marine Applications

Inspect the exhaust manifold slip joint for signs of soot leakage.

If there is no evidence of an exhaust leak from the slip joint and there are no visible cracks or damage, do **not** separate the exhaust manifold sections.



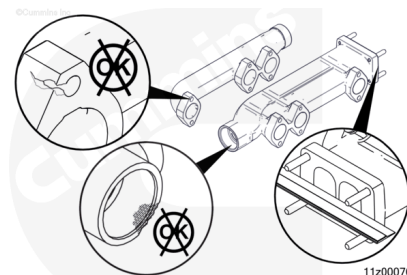
If soot leakage or damage is found, separate the manifold sections.



Inspect the parts for cracks or other damage.

Inspect the exhaust manifold slip joint for wear.

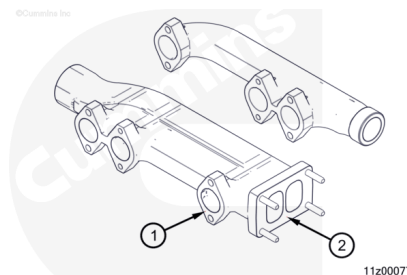
Inspect the exhaust manifold turbocharger mounting surface for warping.



Measure the mounting surfaces for flatness.

The exhaust manifold mounting surface (1) **must** be flat to within 0.25 mm [0.010 in]. The turbocharger mounting flange (2) **must** have a surface flatness of 0.13 mm [0.005 in].

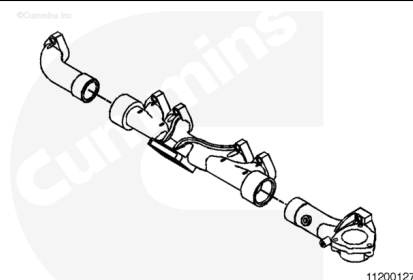
If these surfaces are **not** within the flatness specifications, see the Alternative Repair Manual, L10 and M11 Series Engines, Bulletin 3810310.



Assemble

Automotive with CM875, CM870 and CM570

Install the exhaust manifold end sections to the center section.



Automotive with CM876

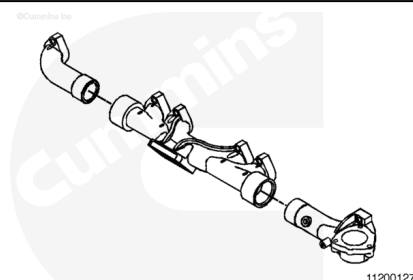
Install the exhaust manifold end sections to the center section.

The end section bolt holes **must** be 225 mm [8.875 inch] center to center before installing onto the cylinder head.

Note : The exhaust manifold sections **must** be assembled on a flat surface for proper alignment.

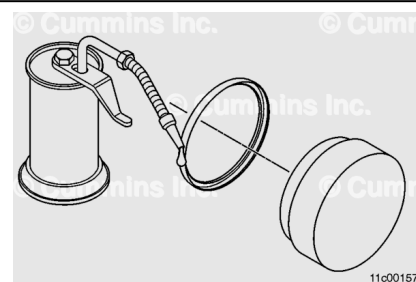
Note : QSM11 engines do **not** have an exhaust manifold seal.

The below steps in the Assemble section do **not** apply to the QSM11 engines.



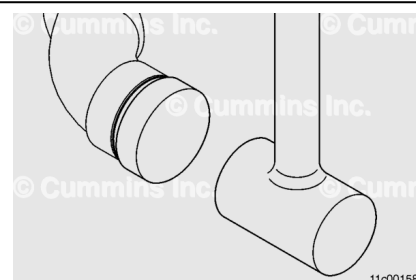
Place the exhaust seal onto the lip of the exhaust seal installer, Part Number 4918709.

Use a light coat of grease or clean engine oil to lubricate the exhaust seal during assembly onto the male and female sections of the exhaust manifold.



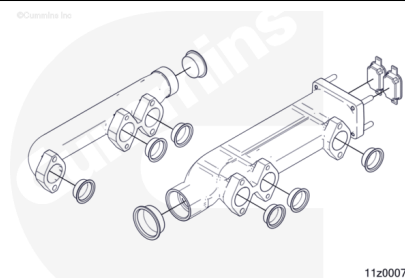
Position the seal installer and seal in line with the sealing surface on the exhaust manifold and strike with a lead or a dead blow hammer until the seal is fully seated.

Seal rings **must** remain square to manifold component axes during installation.

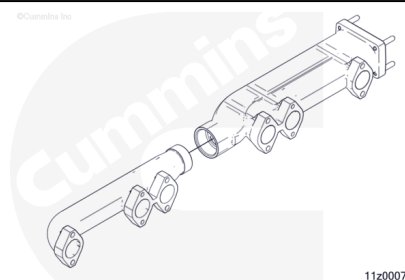


Marine Applications

Remove any protective caps or tape covering the manifold openings.



Assemble the manifold sections.



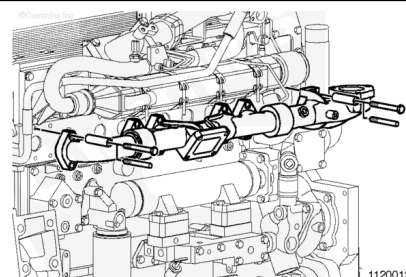
Install

Remove any protective caps or heavy tape.

Install two guide studs, Part Number 3376488, in the two top end holes of the cylinder head exhaust ports.

Install new gaskets between the cylinder head and the exhaust manifold.

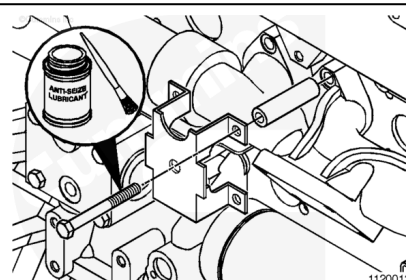
Install the exhaust manifold over the guide pins.



Attach the EGR cooler coolant return tube bracket at the number 4 and number 5 exhaust ports. Orient bracket as shown.

Apply a film of high temperature anti-seize compound to the capscrew threads to be sure of proper loading on the capscrews.

Hand start the capscrews through the bracket, spacers, manifold, and gaskets.

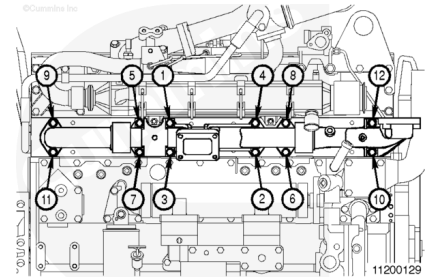


Apply a film of high temperature anti-seize compound to the capscrew threads and install them into the cylinder head.

Remove the guide studs and install the remaining capscrews.

Tighten the capscrews in the sequence shown.

Torque Value: 57 n•m [42 ft-lb]

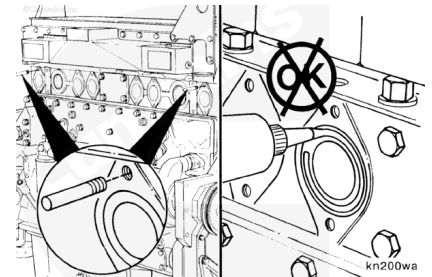


Automotive with CM570

Install two guide studs, Part Number 3376488, in the two top end holes of the cylinder head exhaust ports.

Install a new gasket over each guide stud.

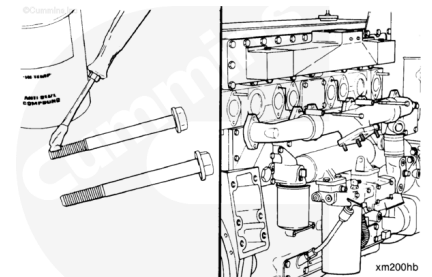
Do **not** use any kind of adhesive to hold the gaskets.



To aid in future capscrew removal, apply a coating of high temperature anti-seize compound to the capscrew threads.

Install the exhaust manifold, tubular spacers, and capscrews.

The exhaust manifold can be tilted back enough to install a new gasket on each exhaust port as the capscrews are being installed.



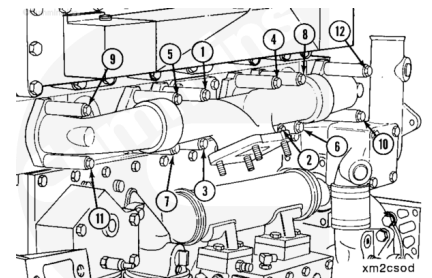
Tighten the capscrews in the sequence shown.

The torque values given have been established using anti-seize compound as a lubricant.

Torque Value:

1. 27 n•m [239 in-lb]

2. 47 n•m [35 ft-lb]



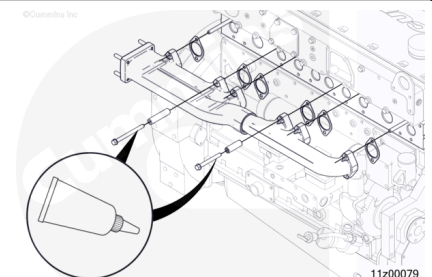
Marine Applications

Apply a film of high temperature anti-seize compound to new exhaust manifold capscrews.

Align the manifold and gaskets using guide studs.

Install spacers and new capscrews hand tight.

Remove guide studs and install remaining spacers and capscrews hand tight.



Tighten the capscrews in the sequence shown.

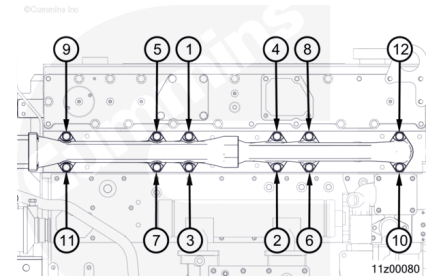
Torque Value:

1. 27 n•m [239 in-lb]
2. 47 Nm 35 ft-lb
3. Loosen all capscrews.

Tighten the capscrews in the sequence shown.

Torque Value:

1. 27 n•m [239 in-lb]
2. 47 N•m 35 ft-lb
3. Loosen all capscrews
4. 27 n•m [239 in-lb]
5. 47 N•m 35 ft-lb.

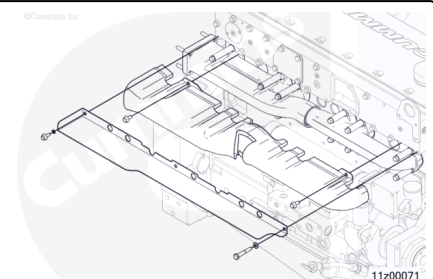


Install the exhaust manifold insulation blanket.

Install the exhaust manifold heat shield.

Tighten the capscrews.

Torque Value: 47 n•m [35 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Automotive with CM570

- Install the turbocharger. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)

- Install the air piping to the turbocharger. See equipment manufacturer service information for the correct torque value.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Operate the engine and check for air or exhaust leaks.

Automotive with CM875, CM870 and CM570

- Install the EGR valve. Refer to Procedure 011-022 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-022-tr.html)
- Install the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)
- Install the water manifold return tube. Refer to Procedure 008-061 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-061-tr.html)
- Install the EGR valve coolant lines. Refer to Procedure 011-030 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-030-tr.html)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)
- Install the exhaust gas pressure sensor tube. Refer to Procedure 011-027 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-027-tr.html)
- Install the air piping to the turbocharger. See equipment manufacturer service information for the correct torque value.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Operate the engine. Check for air or exhaust leaks.

Automotive with CM876

- Install the exhaust gas pressure sensor tube. Refer to Procedure 011-027 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-027-tr.html)
- Install the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)
- Install the water manifold return tube. Refer to Procedure 008-061 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-061-tr.html)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)
- Install the air piping to the turbocharger.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Operate the engine. Check for air or exhaust leaks.

Marine Applications

- Install the heat exchanger. Refer to Procedure 008-053 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-053-tr.html)
- Install the aftercooler assembly (sea water). Refer to Procedure 010-005 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-005-tr.html)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to 70°C [158°F] coolant temperature. Check for leaks.

Last Modified: 15-Jan-2020
